👑 CODERKING STUDIO

Official Technical Roadmaps & Architecture Guides

MASTER ENGINEERING BLUEPRINT

Full-Stack Distributed Systems Roadmap: 1 to 10M Users

The end-to-end architectural patterns to scale web applications with sub-millisecond response and 99.99% uptime.

🚦 1. Tier-1: Ingestion & Edge Gateways

- **Anycast DNS + Cloudflare:** Edge DDoS mitigation and SSL termination.
- **Nginx / Envoy / HAProxy:** Reverse proxy with least-connections load balancing.
- **Rate Limiting:** Token bucket algorithm to protect downstream backend services.

⚡ 2. Tier-2: Caching & Concurrency

- **Redis Single-Threaded RAM:** Sub-ms caching for sessions, counters, and feeds.
- **Bloom Filters:** O(1) probabilistic existence check to prevent Cache Penetration.
- **Cache Invalidation:** Write-Through vs Cache-Aside with strict TTL expiry.

📨 3. Tier-3: Asynchronous Event Streams

- **Apache Kafka / RabbitMQ:** Decouple write operations from slow third-party APIs.
- **Idempotency Keys:** Ensure duplicate payment requests or webhooks are never executed twice.
- **gRPC Microservices:** High-speed binary Protobuf communication between backend clusters.

💾 4. Tier-4: Database & Replication

- **Primary-Replica Sharding:** Write to Primary, read from 3+ Read Replicas.
- **B-Tree & LSM Trees:** Indexing strategies for high-throughput reads vs writes.
- **Consistent Hashing:** Distributed data distribution across multi-node clusters.

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Official GitHub: **github.com/shubh74/coderking-studio**